



WORKSHOP

20th EATEL
Summer School



Burgas, Bulgária



22 May, 2026

Designing Empirical TEL Studies in Higher Education



Workshop Materials:

Website/Slides:

<https://tinyurl.com/jtel26long>

Practical Research Insights
from the CORE Project

Speakers

- Fabíola Gonçalves
- Leon Camus
- Sebastian Gombert
- Gianluca Romano

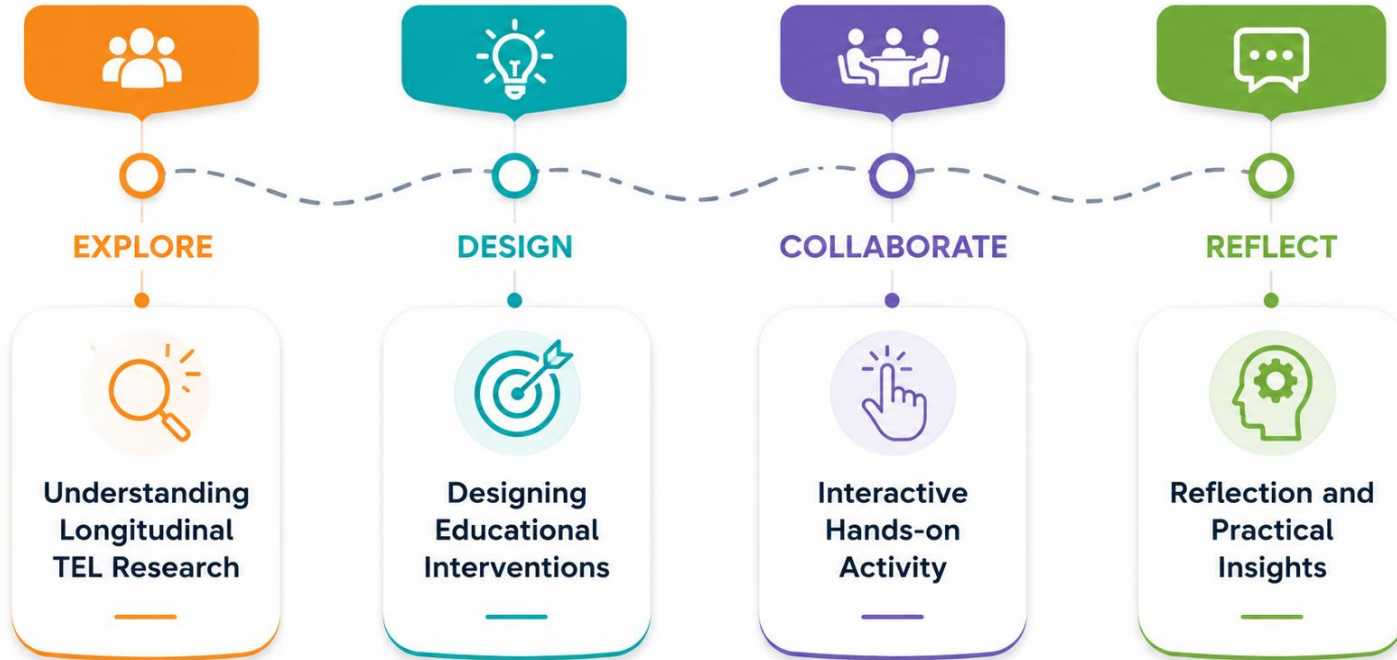


Getting to know each other ...

- ☐ May you introduce yourself?
- ☐ Have you attended other workshops during the Summer School?
- ☐ What do you hope to take away from this workshop?
 - Why have you chosen this workshop?
- ☐ **What have you enjoyed most in the Summer School this year?**
 - What was the most beautiful church that you have visited so far? 😊



Agenda



Critical Online Reasoning (COR)

"Personal abilities of searching, selecting, accessing, processing, and using online information to solve a given problem or build knowledge while critically distinguishing trustworthy from untrustworthy information and reasoning argumentatively based on trustworthy and relevant information from the online environment".

Critical Information Evaluation (CIE)

- ✓ Focuses on students' selection of information sources and evaluation of sources based on online information
 - Ability to differentiate and select high- instead of low-quality information
 - "Cue" (indicator) of meaningful pattern in the online environment of "trust"
 - title or keyword on the search engine results page, an article title, information about author.

Online Information Acquisition (OIA)

- ✓ Focuses on the searching and accessing of online information
 - Use of general and specialized search engines and databases
 - Definition of searching queries, and
 - Selection of specific website (information retrieval)

Reasoning with Evidence, Argumentation, and Synthesis (REAS)

- ✓ Focuses on uniting the initially appraised information, weighting it against further indications and perspectives, and using it as evidence to construct a convincing argument that accounts for uncertainty

Main Facets of COR

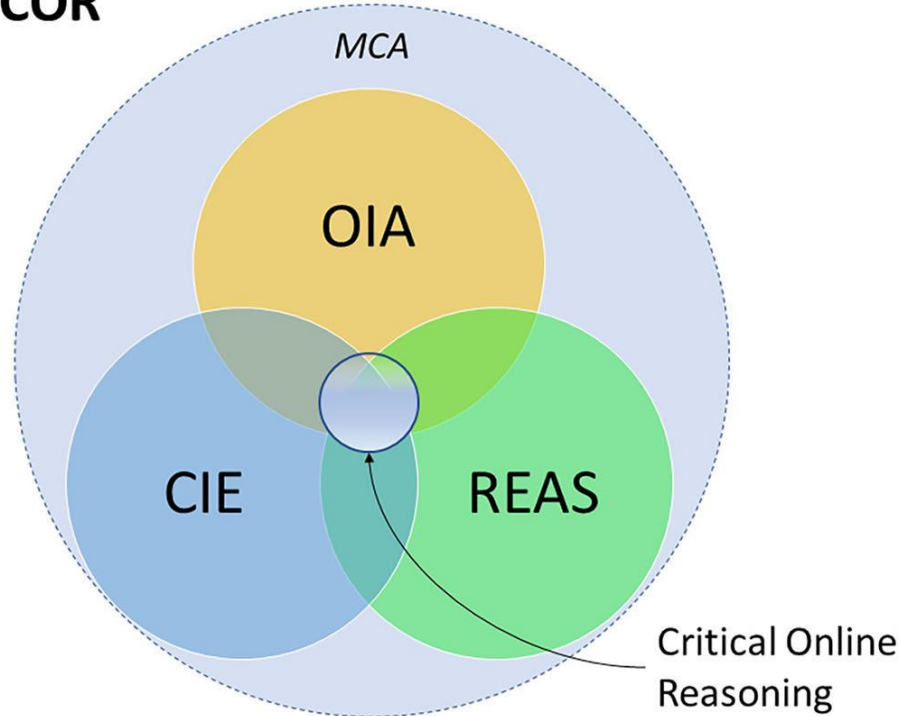
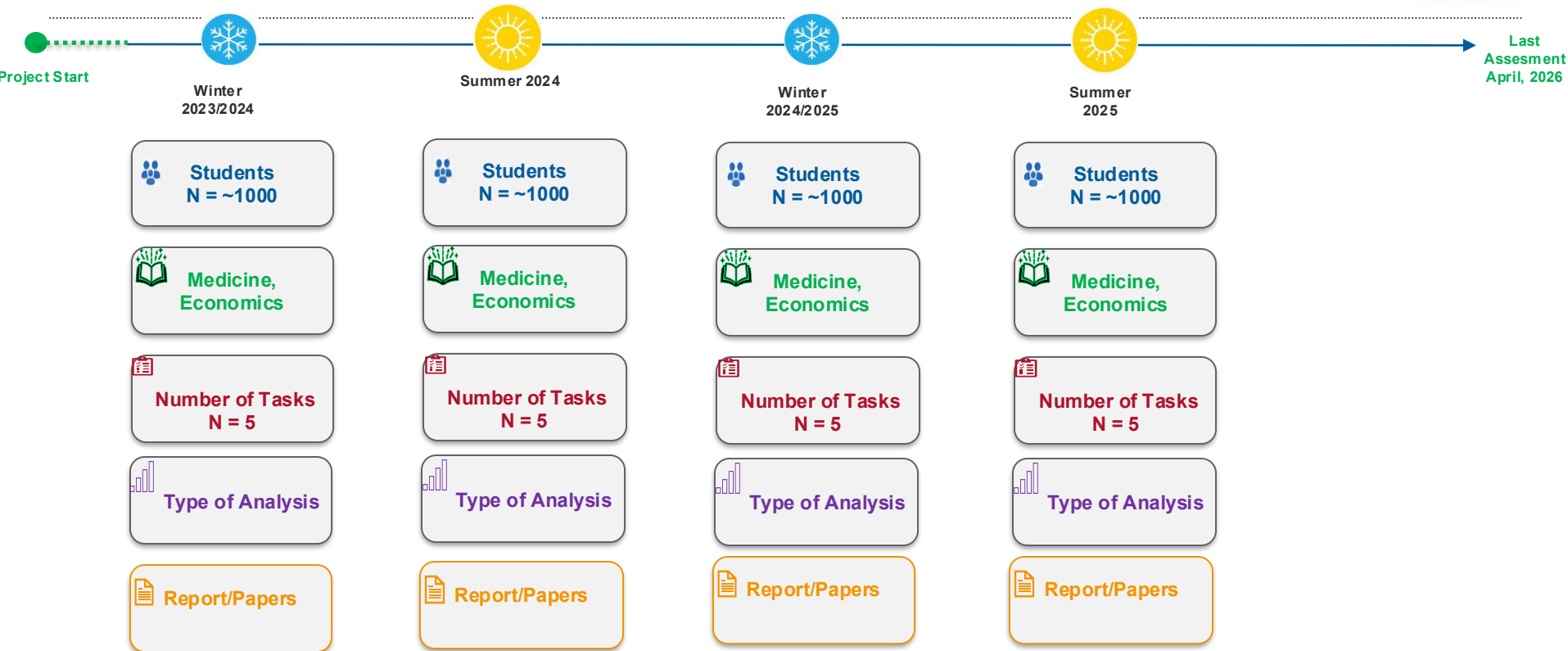


Figure 1: MCA, metacognitive activation; OIA, online information acquisition; CIE, critical information evaluation; REAS, reasoning with evidence, argumentation and synthesis, Zlatkin-Troitschanskaia (2023).

CORE Longitudinal Assessments Overview



Study Focus: Tracing OIA, CIE and REAS over time through longitudinal assessments.

What Come to Mind?

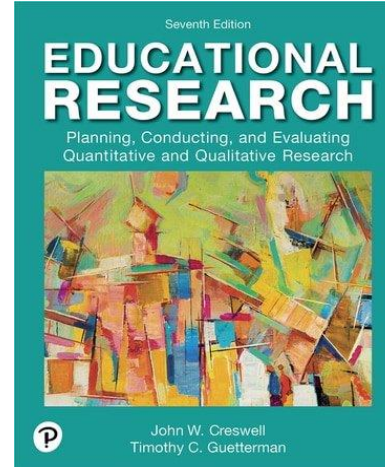
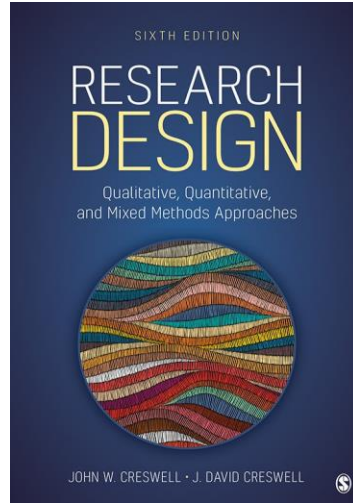
- ☐ What makes a research project *well designed*?
- ☐ Which elements are essential for designing meaningful educational research?
- ☐ In TEL research (or in your own research field), what matters most when designing a study?

Feel free to add multiple responses 



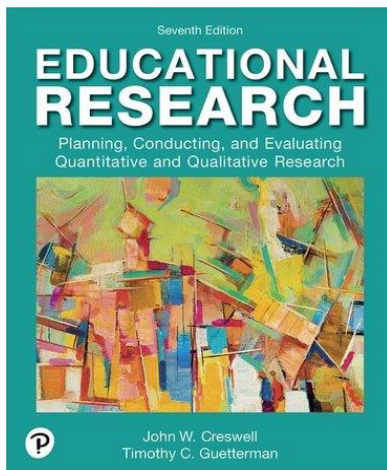
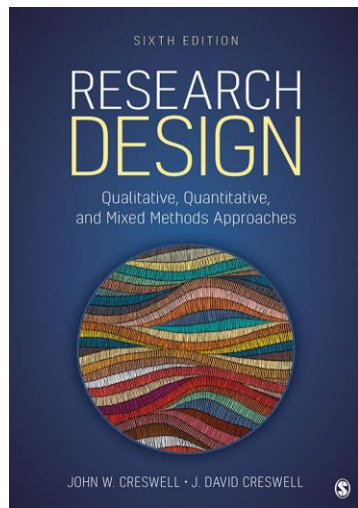
What is Relevant?

- Imagine you are starting a new TEL project today.
 - Where would you begin, which steps would you prioritize first?



- ❑ **Many definitions:**
- ❑ "the systematic study of designing, developing and evaluating educational interventions".
- ❑ "systematic but flexible methodology aimed to improve educational practices through iterative analysis, design, development, and implementation."

Plomp, T., & Nieveen, N. (2013) and Wang, F., & Hannafin, M. J. (2005).



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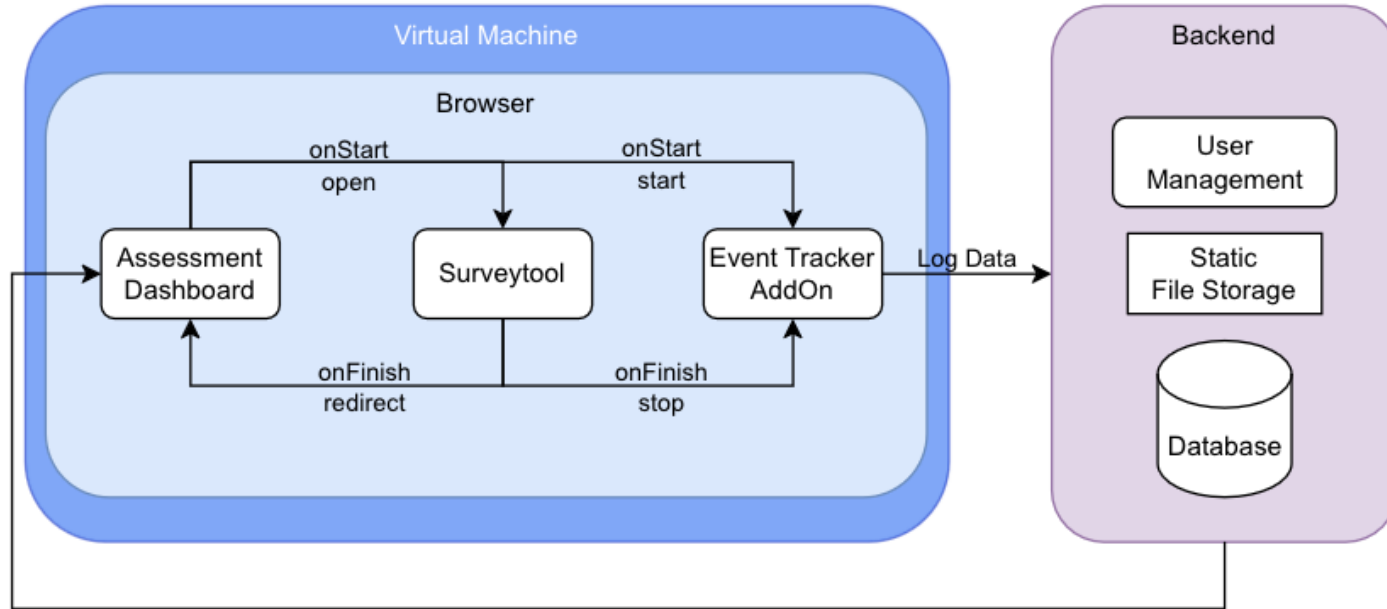
■ Practical Insights From CORE Project

- A well-defined project proposal provided the foundation and guidelines for the longitudinal study.
- The first assessment phase was expected to begin in October 2023.
- The team had approximately three months to organize the overall assessment infrastructure.

■ Main challenges:

- **Short time constraints**
- **Expectations from project leaders**
- **Researchers individual working pace**
- **Coders' ability to deliver their tasks (learning process, available time, and technical expertise)**

CORE: NeoBridge Platform



in einem guten Zustand ist, denkt sie darüber nach, diese für ein Projekt in einem Entwicklungsland zu spenden. Allerdings ist Frau Berg unsicher, wie sinnvoll solche Aktionen sind. Sie bittet daher Dr. Kerzel um Rat, wie solche Projekte zu beurteilen sind. Dieser beauftragt Sie als Praktikant*in schließlich mit einer Internetrecherche zu dem Thema.



Verbleibende Zeit: 29:43

Recherchieren Sie im Internet (mind. 2 Quellen), inwiefern die Spende von nicht länger getragenen Brillen sinnvoll ist. Welche Voraussetzungen sind bei einer Brillenspende jeweils zu beachten? Welche Vor- und Nachteile können im Kontext von Global Health damit verbunden sein? Würden Sie Frau Berg zu- bzw. abraten, ihre Brille zu spenden?

Fassen Sie Ihre Recherchen als Stellungnahme (ca. 150-300 Wörter) zusammen. Stützen Sie Ihre Argumentation auf mindestens 2 der Quellen, die Sie bei Ihrer Suche gefunden haben und geben Sie deren URLs an.

Nehmen Sie sich hierfür insgesamt ca. 15 Minuten Zeit.

0 Wörter

Geben Sie im Folgenden die URLs der für Sie wichtigsten Quellen und den für Sie relevanten Abschnitt/Teil der jeweiligen Quelle an (z. B. [https:// de.wikipedia.org/wiki/Grundgesetz_für_die_Bundesrepublik_Deutschland](https://de.wikipedia.org/wiki/Grundgesetz_für_die_Bundesrepublik_Deutschland); Abschnitt „Parlamentarischer Rat“).

Nehmen Sie sich hierzu ca. 3 Minuten Zeit.

NeoBridge Data Dashboard

author@institute.com

- Dashboard
- Attempts
- Manage API Keys
- API Specification
- (x) Variable Storage

Attempts

BY ASSESSMENTS BY TASKS BY PARTICIPANTS BY FINISHED STATUS REFRESH

CLOSE OR REOPEN ATTEMPT RENDER AS VIDEO DOWNLOAD

☐	Id	Participant	Assessment ↑	Assessment Step	Start Time	End Time
☐	42334	pxe8c4	SoSe 25 GEN KI	Nudging	2025-07-06T23:17:09.301321Z	2025-07-06T23:58:07.257725Z
☐	42332	23qg7a	SoSe 25 GEN KI	Spinne	2025-07-06T22:27:33.677774Z	2025-07-06T22:34:39.989423Z
☐	42331	23qg7a	SoSe 25 GEN KI	Windpark	2025-07-06T22:11:07.759866Z	2025-07-06T22:27:30.372463Z
☐	42330	23qg7a	SoSe 25 GEN KI	GEN KI	2025-07-06T22:06:44.543461Z	2025-07-06T22:11:02.470449Z
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☐	42328	4ds71u	SoSe 25 GEN KI	Windpark	2025-07-06T21:40:47.614430Z	2025-07-06T21:51:16.507154Z
☐	42327	4ds71u	SoSe 25 GEN KI	Mount Everest	2025-07-06T21:23:24.146887Z	2025-07-06T21:40:45.283360Z
☐	42326	4ds71u	SoSe 25 GEN KI	GEN KI	2025-07-06T21:21:13.583919Z	2025-07-06T21:23:21.880867Z
☐	42325	4ds71u	SoSe 25 GEN KI	Vorbefragung	2025-07-06T21:11:10.693292Z	2025-07-06T21:21:11.906502Z
☐	42324	3qtwa3	SoSe 25 GEN KI	Startup	2025-07-06T20:49:24.144363Z	2025-07-06T21:00:03.846961Z
☐	42323	5m3sk4	SoSe 25 GEN KI	Pilotenstreik	2025-07-06T20:39:17.190163Z	2025-07-06T21:40:07.395736Z

1-1050 of 1050

AG	AH	AI	AJ
id	inputName	inputType	logEntryType
443645	-	-	PageFocus
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15313555	-	-	Screenshot
120498	-	-	TabFocus
102122	-	-	TabOpen
4986477	-	-	PageTextState
102123	-	-	TabOpen
274586	-	-	PageOpen
274030	-	-	PageClose
443646	-	-	PageFocus
274587	-	-	PageOpen
15313556	-	-	Screenshot
274585	-	-	PageOpen
4986481	-	-	PageTextState
4986483	-	-	PageTextState
7939914	IT21_01	radio	InputChange
15313557	-	-	Screenshot
7939915	IT22	radio	InputChange
15313558	-	-	Screenshot
44011	-	-	ButtonClick
274031	-	-	PageClose
443648	-	-	PageFocus
274588	-	-	PageOpen
15313574	-	-	Screenshot
4986497	-	-	PageTextState
13503401	-	-	TextMarked
7939919	IT17_01	radio	InputChange

Log Event	Description
Text Marked	Logged whenever text is selected or the active selection changes; stores the selected text.
Input Change	Logged when user input changes in text inputs, textareas, or contentEditable elements (excluding password fields); stores the entered content.
Keystroke	Logged for each key press (excluding password fields); stores the key code.
Copy	Logged when text is copied; stores clipboard content.
Paste	Logged when text is pasted; stores pasted content.
Button Click	Logged when a button or input of type button or submit is clicked; stores button captions.
Link Click	Logged when a hyperlink is clicked; stores URL and link captions.
Scroll	Logged when scrolling occurs (max. every 200 ms); stores absolute and relative X/Y coordinates.
Viewport	Logged when the viewport changes (max. every 200 ms); stores coordinates as well as tab width and height.
HTML State	Logged on page load and whenever HTML changes; stores raw HTML compressed via gzip.
Page Text State	Logged on page load and whenever text changes; cleaned via readability.js and stored gzip-compressed.
Mouse Movement	Logged when cursor position changes (max. every 200 ms); stores relative X/Y coordinates.
Screenshot	Logged with most other events (excluding Copy, Paste, Button/Link Click, and window/tab events; max. every 200 ms); captures viewport as gzip-compressed PNG. Timestamped sequences allow video reconstruction.

Research Design of an Interactive Empirical Study

- ❑ Please access the following link:

Strawberries & Exam Performance
A Longitudinal Study



Breathing Exercise and Academic Performance
A Longitudinal Study



**Hands-
on task:**

- ❑ Let's design a TEL intervention together?

☐ Let's design a TEL intervention together?

☐ Workshop Materials & Longitudinal TEL Examples

- ☐ Scan the QR code to explore:
 - ☐ GitHub workshop materials:
 - ☐ **Breathing Exercise & Academic Performance**
 - ☐ **Strawberries & Exam Performance**



<https://tinyurl.com/jtel26long>

**Hands-on
task:**

Workshop Evaluation



1. Creswell, J. W., & Creswell, J. D. (2018). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches* (5th ed.). SAGE
2. Plomp, T., & Nieveen, N. (Eds.). (2013). *Educational Design Research: Part A. An Introduction*. Enschede, The Netherlands: Netherlands Institute for Curriculum Development (SLO).
3. Zlatkin-Troitschanskaia, Olga & Fischer, Jennifer & Molerov, Dimitri. (2023). Critical Online Reasoning in Higher Education (CORE) – Overview of the Research Unit: Objectives, framework, structure, and research projects (CORE Working Papers, 1).
4. Wang, F., & Hannafin, M. J. (2005). Design-based research and technology-enhanced learning environments. *Educational Technology Research and Development*, 53(4), 5–23.